Clustering of Credit Card Customer Data

Results and Analysis

Overview

This report presents a comparative evaluation of three unsupervised machine learning algorithms: K-Means, Hierarchical Clustering, and Density-Based Spatial Clustering of Applications with Noise (DBSCAN) which were applied to the Credit Card dataset. The objective is to assess the effectiveness of each algorithm in identifying meaningful groupings within the data and determine the most suitable approach for the clustering task.

The algorithms are compared based on cluster compactness and separation, using the Silhouette Score where applicable, alongside visual assessments of cluster quality and computational efficiency. These evaluation criteria provide complementary perspectives on the structure of the resulting clusters, the distinction between groups, and the processing requirements of each algorithm.

Based on the comparative results, the report identifies the most suitable clustering algorithm and discusses the factors supporting its selection. Particular consideration is given to the characteristics of the resulting clusters and each algorithm's ability to identify meaningful patterns in the dataset. The report concludes by summarizing the key findings, the strengths and limitations observed during the evaluation, and the rationale for selecting the final algorithm.

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1. Model Comparison

1.1 Cluster Compactness and Separation

Table 1: Model Comparison Summary

Model	Silhouette Score (%)	Visual Quality of Clusters	Computational Efficiency (Processing Time)
K-Means	16.2	Poor	Fastest
Hierarchical Clustering	26.7	Weak to Moderate	Slowest
DBSCAN	27.8	Slightly Better	Moderate

a. Silhouette Score

All three algorithms produced relatively low silhouette scores, indicating that the dataset does not exhibit strong natural clusters. K-Means achieved the lowest score (16.2%), showing poor cluster separation and cohesion, likely due to varying cluster densities and non-spherical shapes in the data. Hierarchical Clustering performed moderately (26.7%) but still revealed overlapping clusters with weak separation. DBSCAN achieved the highest score (27.8%), suggesting it captured slightly better-defined density-based structures, although overall cluster quality remained weak across all methods.

b. Visual Quality of Clusters

The visual inspection of the PCA scatter plots provides additional insight into how well each clustering algorithm separates the data in two-dimensional space.

- K-Means produced the weakest visual separation.
  Clusters largely overlapped, and boundaries were not clearly defined.
  This aligns with its low silhouette score (16.2%), suggesting that the data does not naturally form spherical, centroid-based clusters appropriate for K-Means.
- Hierarchical Clustering showed slightly improved structure.
  Although clusters were still overlapping, the dendrogram and PCA plot revealed

some weak subgrouping patterns.

This is consistent with its moderate silhouette score (26.7%), indicating partial but still limited cluster separation.

- DBSCAN provided the best visual separation among the three models.

  It identified a few distinct dense regions and isolated some noise points, producing marginally clearer cluster boundaries than the other methods.

  This matches its highest silhouette score (27.8%), although the improvement remains modest.

Overall, the visual plots confirm the quantitative scores: all three algorithms show weak separation, with DBSCAN performing slightly better and K-Means performing worst.

c. Computational Efficiency

Processing time varied noticeably across the three algorithms due to differences in their underlying computational complexity.

- **K-Means was the fastest model.**

  It converged quickly because it does not compute a distance matrix and operates with low computational cost per iteration.

  Its efficiency makes it suitable for large datasets, although its accuracy was poor for this task.

- **Hierarchical Clustering was the slowest.**

  The method requires constructing a full distance matrix ($O(n^2)$), which significantly increases execution time and memory usage.

  This trade-off is typical of hierarchical methods, which provide structural insights through dendrograms but are computationally expensive.

- **DBSCAN showed moderate processing time.**

  Its performance depends on neighborhood searches (ε-radius), making it slower than K-Means but still faster than Hierarchical Clustering.

  It offers a balance between efficiency and the ability to detect arbitrarily shaped clusters.

Overall, the computational ranking is: K-Means (fastest) > DBSCAN (moderate) > Hierarchical Clustering (slowest).

2. Best Algorithm

Based on the overall evaluation, DBSCAN emerges as the most suitable algorithm for this dataset, although its advantage is only modest. It achieved the highest silhouette score (27.8%), indicating slightly better cluster separation compared to K-Means and Hierarchical Clustering. The PCA visualization also showed that DBSCAN was able to identify a few distinct dense regions and isolate noise points, resulting in clearer boundaries than the other methods. Its computational efficiency is reasonable, and its ability to detect non-spherical clusters makes it better aligned with the underlying structure of the data. However, the overall cluster quality across all algorithms remains relatively weak, suggesting that the dataset does not exhibit strong natural groupings. Despite this, DBSCAN provides the most meaningful clustering result among the three methods.

3. Conclusion

Overall, the dataset shows weak natural clustering structure across all methods, as reflected by consistently low silhouette scores and overlapping clusters in visualizations. Among the three algorithms, DBSCAN provides the most meaningful results, offering slightly better separation, clearer visual patterns, and reasonable computational performance. K-Means performs the poorest due to its sensitivity to cluster shape and density, while Hierarchical Clustering provides moderate structure but suffers from high computational cost and limited separation. Therefore, DBSCAN is the most appropriate algorithm for this dataset, although further tuning and dimensionality reduction may be required to achieve stronger clustering performance.